

Merits and Demerits of Setting up Smart Grids

With more than 1.2 billion people in India and 230GW of installed capacity, any small improvement in the utility will have a magnified effect when it comes to gains. Smart grid technologies are vital to meeting India's deficit in electrical power. Let us see where we are with its implementation

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In a power-starved country like India, technologies that can revolutionise the power grid may not have arrived at a better time. People, who feel that smart grid technologies only help those who already have access to basic electrical power, are mistaken. Not only they modernise our aging grids but they also help move towards more efficient power transmission, thus making more power available for distribution.

Fourteen smart grid pilot projects were approved for 50 per cent fund-

ing by the government of India in July 2013. The Smart Grid Vision and Roadmap document for India, a joint document by Ministry of Power, India Smart Grid Task Force and India Smart Grid Forum, was also released.

Out of the total 230GW capacity, the private sector contributes 73MW with approximately 12 per cent coming from renewable energy sources.

Major factors driving growth

When we talk about fast-moving markets, energy is a big one. Smart grid is a really big investment area around the world where the engineers need to

move fast, but the requirements are not really specified yet. Thus there is a lot of opportunity for innovative solutions here. The Ministry of Power's Smart Grid Vision and Roadmap for India, released in the second half of 2013, already places highest importance to providing supply of electricity to all households by 2017 through smart grid technologies. Most of these only implement basic building blocks required for smart grids, so the infrastructure can be later leveraged to transform them using smart grid technologies.

Reji Kumar Pillai, president of the India Smart Grid Forum (ISGF), states in the roadmap preface that, "India can build smart grids with little or marginal costs as we have much better business cases for the same from huge efficiency gains that can accrue through automation and elimination of wastages of all kinds." He adds that the market size of smart grid projects envisioned in the smart grid vision roadmap is about 30-40 billion dollars in the next 15-20 years.

Manufacturing in India has also started to see an upturn. Avantha Group Company CG launched its state-of-the-art smart grid facility at the Global Village in Bangalore on January 13, 2014, for full-fledged manufacturing of smart grid devices. Besides manufacturing smart grid devices, the facility will support economic development, foster job creation and boost an

